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Environmental pollutant and source analysis near Chinese chemical industrial activities

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Abstract
This study used LC-QTOF-MS and suspect screening. It systematically identified and analyzed pollutants in the rivers and coastal areas around a chemical industrial park in Jiangsu Province. The research revealed the spatial distribution of complex chemical pollutants and their links to industrial discharges. Samples from the GH, ZB, and XY rivers and coastal areas revealed a complex mix of pollutants, including pesticides, pharmaceuticals, and industrial chemicals. PCA traced pollution sources, with industrial waste and agricultural runoff being significant. The study identified 120 chemicals, with pesticides making up 65%. Notable contaminants included nonylphenol and tributylamine. Industrial contributions were evident downstream of the park, with sites G-3 to G-6 in the GH River and ZB-4 in the ZB River showing high contamination. PCA identified industrial discharges and agricultural runoff as primary pollution sources, with chemicals like anileridine, atrazine, and azoxystrobin indicating industrial influence. Unique pollution signatures suggested localized impacts. The findings provide a scientific basis for developing targeted pollution control strategies and promoting sustainable development of chemical industrial parks.

Keywords Suspect screening · Hazardous chemicals · Chemical industrial park · Rivers · Wastewater treatment plants

1 Introduction

Since the 1990s, the chemical industry has increasingly aggregated into industrial zones, a trend particularly pronounced in China with the establishment of chemical industrial parks (CIPs). The number of national key CIPs has surged from 200 in 2009 to 676 in 2018, highlighting their strategic importance (China Petroleum and Chemical Industry Federation, http://www.cpcia.org.cn/). These parks offer several advantages, including efficient infrastructure utilization, enhanced industrial collaboration, reduced operational costs, and streamlined governance and regulatory oversight. Consequently, nearly 45% of large chemical enterprises now operate within these parks.

However, Chemical Industrial Parks have become significant sources of environmental pollution, with pollutants detected in air, water, and soil [10, 13, 21]. Incidents like chemical leaks have adversely affected local health [11, 40, 42, 44]. Current wastewater treatment focuses on traditional pollutants, often overlooking emerging contaminants [8, 12].

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This highlights the need for comprehensive policies and innovative technologies to address complex pollution issues. Compounds such as pharmaceuticals, active pharmaceutical ingredients, persistent organic pollutants (POPs, 45; [26, 27]), and personal care products are challenging to remove from wastewater, underscoring the importance of developing more effective treatment solutions to safeguard environmental and public health [9, 18].

Recent research has further highlighted the importance of suspect screening in identifying and managing emerging contaminants. Manz et al. [23] provided a comprehensive review of non-targeted analysis (NTA) and suspect screening analysis (SSA) in examining the chemical exposome, emphasizing their role in detecting unknown or suspicious chemical substances in environmental and biological samples. This approach has been successfully applied in various studies, including the detection of chemical residues in aquacultured products [36] and the assessment of household cleaning products [25]. These studies underscore the potential health and environmental risks associated with the presence of these chemicals.

Pollution source identification is crucial for effective environmental management, utilizing techniques such as chemical fingerprinting, isotope analysis, and modeling. Suspect screening, a non-targeted analytical approach, plays a pivotal role in identifying a broad spectrum of potential pollutants, thereby aiding in source tracking. This method provides comprehensive pollutant coverage, early detection of emerging contaminants, and improved regulatory efficiency through targeted mitigation strategies [32]. Technological advancements in non-target and suspect screening, particularly the integration of high-resolution mass spectrometry (HRMS) with chromatography, have transformed the detection of elusive pollutants [32], [45]). These techniques have proven indispensable in identifying unknown pollutants in wastewater and surface water, significantly enhancing our ability to monitor and assess their environmental impacts [16, 37].

Building on prior research in Jiangsu Province's CIPs, where suspect screening and liquid chromatography time-of-flight mass spectrometry (LC-QTOF-MS) were employed to identify pollutants across different industrial segments and wastewater treatment stages [20], this study extends the investigation to rivers adjacent to a CIP, including downstream tributaries and coastal estuaries. Unlike Liu et al. [20], which focused primarily on wastewater within the CIP, this study explores the correlation between pollutants in the water and industrial discharges in a broader environmental context. This comprehensive screening not only deepens our understanding of the environmental repercussions of industrial emissions but also underscores the imperative for enhanced wastewater treatment and environmental monitoring within industrial settings.

2 Materials and methods

2.1 Sample collection

The study involved collecting 29 surface water samples in wet season (in July) from various rivers around the CIPs in Jiangsu Province, including the GH River, ZB River (which runs through the CIP), XY River, and the Yellow Sea coast. The samples were collected from different sites along each river, with specific sites denoted based on their location relative to the CIP and other landmarks. Samples were also collected from sites near the GH River estuary at the coast.

Each sample collected was 1 L in volume and underwent filtration through 1 μm glass fiber filter membranes and 0.45 μm nylon filter membranes to remove solid impurities. The filtered samples were then stored at 4 °C and transported to the laboratory for further analysis. This meticulous sampling process was conducted to investigate the presence of pollutants in the rivers surrounding the CIP and to assess any potential relationships between water quality and industrial discharges.

2.2 Sample pretreatment

The pollutants present in the water samples were extracted using solid phase extraction (SPE; [20]). A preconditioned Oasis HLB cartridge (6 cc/500 mg, Waters, USA) was used for the extraction process. The extraction flow rate was maintained at 3–5 mL/min, and after extraction, elution was carried out using 10 mL of methanol followed by 10 mL of dichloromethane (Merck, Germany). The extracted sample was then evaporated using N_2 and dissolved in 1 mL of methanol. The resulting extract was filtered through a 0.2 μm polypropylene membrane syringe filter (Acrodisc® GHP, 13 mm, 0.2 μm, Waters, USA), collected in an amber vial, and stored at − 20 °C until further analysis.

Prior to the extraction procedure, all apparatus underwent rigorous cleaning using hexane, dichloromethane, and methanol to eliminate any potential contamination. This careful preparation guaranteed the purity of the water samples, enabling accurate pollutant analysis in subsequent stages.

2.3 LC-QTOF-MS analysis and mass spectrometry data analysis

This research employed the LC-QTOF-MS technique, following the same analytical protocol established in our earlier work [20]. Briefly, the samples underwent analysis through liquid chromatography (LC,Agilent Technologies, Waldbronn, Germany) integrated with a high-resolution hybrid quadrupole time-of-flight mass spectrometer (Triple TOF 5600, AB Sciex, Foster City, CA). Comprehensive details regarding instrument configurations, quality assurance measures, and quality control procedures are included in the supplementary materials (SI).

The suspect peaks were separated and pollutants were characterized by referencing standard compounds and databases (see supplementary information for details). The identification process followed established methodologies, with each pollutant assigned a confidence level based on Schymanski et al. [31]. A confidence level of 1 was assigned when the structure was verified using authentic standards, while level 2 indicated a probable structure, either through spectral library matching (2a) or diagnostic evidence (2b). Only compounds with confidence levels of 1 or 2 were included in this investigation.

2.4 Data analysis tools

The mass spectrometry data were processed using PeakView 2.2 (AB SCIEX, USA). For multivariate analysis, principal component analysis (PCA) was conducted with SPSS Statistics version 24.0 (IBM, USA). Graphical representations were generated using SigmaPlot 12.5 (Systat, USA), Excel 2016 (Microsoft, USA), and Venny 2.1 (available at https://bioinfogp.cnb.csic.es/tools/venny/index.html).

3 Results and discussion

3.1 Pollutant identification in rivers

3.1.1 Pollutant identification in the GH river

The suspect screening approach revealed 55 unique chemical compounds in the GH River, with comprehensive data presented in Table S4. As depicted in Fig. 1a, both the overall response intensity and the number of detected contaminants exhibited notable differences across sampling locations, highlighting substantial spatial heterogeneity. Notably, sites G-3 to G-6 recorded substantially higher response intensities than the upstream sites G-1 and G-2, with site G-6 showing a response intensity tenfold that of G-2. This elevated pollution is likely influenced by the proximity of these sites to the CIPs, located between G-4 and G-5. The spatial distribution of pollutants suggests that industrial discharges from the CIP are a major contributor to the pollution levels observed in these downstream sites (Zhao et al., [46]; Lee et al., [47]; Chen [6]).

A consistent trend in pollutant detection and response intensity was observed across all sites from G-1 to G-10. The detected chemicals were categorized by their application, as depicted in Fig. 2. The classification includes 8 bactericides, 3 plasticizers, 1 dye, 12 herbicides, 5 insecticides, 11 pharmaceuticals, 7 intermediates for pharmaceuticals or dyes, 1 pesticide, and 7 chemicals of miscellaneous use. Pesticides, which encompass bactericides, herbicides, insecticides, and plant growth regulators, accounted for 47.3% of the total identified chemicals. The high proportion of pesticides suggests that agricultural activities in the surrounding areas may also contribute to the pollution load in the GH River (Smith, et al., [49]; Zhang et al., [50]).

Moreover, 13 of the detected substances are recognized as hazardous chemicals in China, as listed in Table S4. The presence of these substances underscores the environmental and public health risks they pose. For instance, several of the identified hazardous chemicals, such as certain pharmaceuticals and pesticides, have been reported to have endocrine-disrupting effects and potential carcinogenic properties (Hatch et al., [52]; Vandenberg et al., [53]; Zoeller, [54]). Therefore, ongoing monitoring and robust management strategies are essential to mitigate their impact on the river's ecosystem and adjacent communities. This includes the implementation of stricter discharge regulations

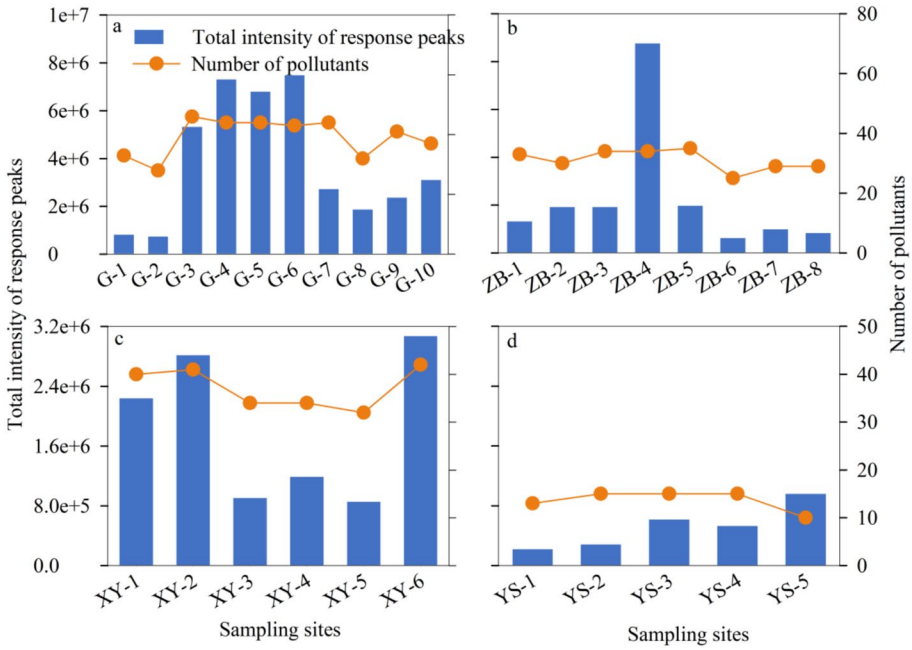

Fig. 1 Total intensity of the response peaks and the number of pollutants in the (**a**) GH River, (**b**) ZB River, (**c**) XY River, and (**d**) YS. River flow direction: G-1 → G-2 → … → G-10 → GH River, ZB-1 → ZB-2 → ZB-3 → ZB-4 → GH River, ZB-8 → ZB-7 → ZB-6 → ZB-5 → GH River, XY-1 → XY-2 → GH river, XY-3 → XY-4 → GH river, XY-5 → XY-6 → GH river

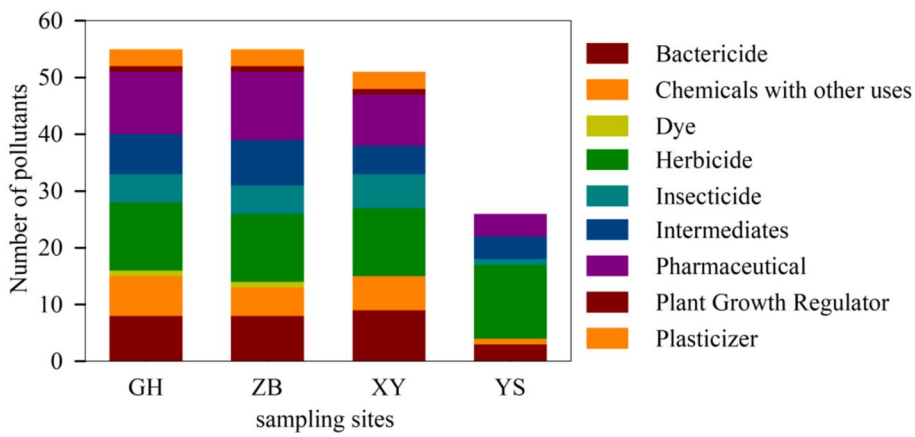

Fig. 2 Classification of pollutants identified by their use

for industrial and agricultural activities, as well as the development of advanced wastewater treatment technologies to remove these hazardous chemicals more effectively (Li et al., [55]; Rodriguez et al., [56]).

A thorough analysis of these chemical profiles and their potential sources is imperative for informing targeted remediation measures. By identifying the specific sources of pollution, such as industrial discharges or agricultural runoff, more effective strategies can be developed to reduce the release of these chemicals into the environment. For example, the use of green chemistry principles in industrial processes and the promotion of sustainable agricultural practices can help minimize the generation and release of hazardous chemicals (Lancaster et al., 2016; Zhang et al., 2020). Additionally, the application of advanced analytical techniques, such as HRMS and liquid chromatography time-of-flight mass spectrometry (LC-QTOF-MS), can enhance the detection and identification of emerging contaminants, providing valuable information for regulatory and remediation efforts (Teng et al., 2021; Li et al., [59]).

3.1.2 Pollutant identification in the ZB river

Our suspect screening analysis of the ZB River samples, detailed in Table S5, identified the presence of 55 chemical substances. Figure 1b illustrates the total response intensity and the quantity of pollutants detected at each sampling site. Notably, site ZB-4 exhibited a response intensity 13 times higher than that of site ZB-6. This significant difference suggests that ZB-4, located within the CIPs, is heavily influenced by pollution. This finding is consistent with previous studies that have shown industrial parks to be major sources of chemical pollution in adjacent water bodies (Smith et al., 2015). The elevated pollution levels at ZB-4 necessitate a deeper investigation into the specific contamination sources and their ecological impacts, particularly given the potential for these pollutants to bioaccumulate and affect aquatic life and human health [43].

The 55 detected substances were categorized by their application, as shown in Fig. 2. The classification includes: 12 pharmaceuticals, 12 herbicides, 8 bactericides, 1 dye, 5 insecticides, 3 plasticizers, 8 intermediates for pesticides or dyes, 1 plant growth regulator, and 5 chemicals of other uses. This diverse chemical profile highlights the multifaceted nature of pollution impacting the ZB River. The presence of a wide range of chemical substances suggests multiple pollution sources, including industrial discharges, agricultural runoff, and possibly domestic wastewater (Liu et al., 2023).

Of particular concern, 15 of these substances are recognized as hazardous chemicals in China (Table S5), posing significant risks to both aquatic ecosystems and human health. These hazardous chemicals include several pharmaceuticals and pesticides, which have been reported to have endocrine-disrupting effects and potential carcinogenic properties (Zhang et al., [63]). The presence of such chemicals underscores the need for stringent regulatory measures to control their release into the environment. Previous studies have shown that even low concentrations of these substances can have adverse effects on aquatic organisms and human health (Kim et al., [64].

These findings underscore the necessity for persistent monitoring and strategic interventions to safeguard the aquatic environment and public well-being. Continuous monitoring can provide valuable data on pollution trends and help identify emerging contaminants, while strategic interventions can target specific sources of pollution (Chen et al., 2022). To effectively control pollution, detailed chemical profiling and source tracing of these contaminants are imperative. Advanced analytical techniques, such as HRMS and liquid chromatography time-of-flight mass spectrometry (LC-QTOF-MS), can enhance the detection and identification of emerging contaminants, providing valuable information for regulatory and remediation efforts (Li et al., 2019; Teng et al., 2021). Such efforts are crucial for developing targeted pollution mitigation strategies aimed at preserving the long-term health of the ZB River and its surrounding areas [43]. This includes the implementation of stricter discharge regulations for industrial and agricultural activities (Lancaster et al., 2016, Smith et al., 2015), as well as the development of advanced wastewater treatment technologies to remove these hazardous chemicals more effectively (Zhang et al., 2020). Furthermore, coordinated efforts from regulatory authorities, local industries, and scientific researchers are essential to combat the detrimental effects of these pollutants. Collaboration between these stakeholders can facilitate the sharing of data, resources, and expertise, leading to more effective pollution control measures (Chen et al., [65]).

3.1.3 Pollutant identification in the XY River

The upper reaches of the XY River, situated above its confluence with the GH River, were divided into three sections—north, middle, and south—with sluice gates installed at each segment. A comparative analysis of suspect screening results between sluice gate sites (XY-1, XY-3, XY-5) and those outside (XY-2, XY-4, XY-6) revealed the presence of 51 chemical substances in the XY River samples, as detailed in Table S6. Figure 1c illustrates the total response intensity and quantity of pollutants detected, with lower intensities observed within the sluice gates. This reduction in pollutant inflow suggests that sluice management plays a pivotal role in mitigating pollution levels in riverine systems. The effectiveness of sluice gates in controlling pollution inflow has been demonstrated in previous studies, highlighting their importance as a physical barrier to reduce contaminant transport (Smith et al., 2015).

The detected substances were categorized by application, with 54.9% identified as pesticides, reflecting the substantial agricultural influence on the river's chemical composition. The high proportion of pesticides indicates that agricultural runoff is a significant contributor to the pollution load in the XY River. Previous studies have shown that pesticides can persist in aquatic environments and have adverse effects on non-target organisms, including aquatic plants, invertebrates, and fish [33], Karadag et al., [17]). The presence of these chemicals underscores the need for sustainable agricultural practices to minimize the release of pesticides into water bodies. This includes the adoption of integrated pest management (IPM) strategies, which can reduce pesticide use while maintaining agricultural productivity (Zhang et al., 2020).

14 substances were identified as hazardous chemicals in China, as listed in Table S6, posing significant environmental and health risks. These hazardous chemicals include several pesticides and pharmaceuticals, which have been reported to have endocrine-disrupting effects and potential carcinogenic properties (Vandenberg et al., 2014; Rodriguez et al., 2018). The presence of such chemicals in the XY River highlights the need for stringent regulatory measures to control their release into the environment. Previous studies have shown that even low concentrations of these substances can have adverse effects on aquatic organisms and human health (Hatch et al., 2013; Zoeller et al., 2014).

Continuous monitoring and regulatory measures are imperative to understand the sources and long-term ecological impacts of these chemicals and to safeguard the river and its habitats. The implementation of stricter discharge regulations for agricultural and industrial activities, as well as the development of advanced wastewater treatment technologies, can help reduce the release of hazardous chemicals into the environment (Lancaster et al., 2016; Li et al., 2021).

Furthermore, coordinated efforts from regulatory authorities, local industries, and scientific researchers are essential to combat the detrimental effects of these pollutants. Collaboration between these stakeholders can facilitate the sharing of data, resources, and expertise, leading to more effective pollution control measures (Smith et al., 2015; Lancaster et al., 2016). For example, the use of green chemistry principles in industrial processes and the promotion of sustainable agricultural practices can help minimize the generation and release of hazardous chemicals (Zhang et al., 2020; Lancaster et al., 2016).

In conclusion, the identification of 51 distinct chemicals in the upper reaches of the XY River, including 14 hazardous substances, highlights the complex pollution challenges facing this water body. The significant proportion of pesticides and the presence of hazardous chemicals underscore the need for comprehensive environmental management strategies (Smith et al., 2015; Hatch et al., 2013). Through ongoing monitoring, detailed chemical profiling, and coordinated efforts from multiple stakeholders, it is possible to mitigate the environmental and public health risks associated with these pollutants and protect the long-term health of the XY River ecosystem (Chen et al., 2022; Li et al., 2021).

3.1.4 Coastal suspect screening at YS sites

Coastal suspect screening at the YS sites detected 26 chemical substances, with response intensities and pollutant quantities detailed in Table S7 and Fig. 1d. The elevated response intensity at YS-5, a site distant from the river estuary, may be attributed to water flow direction or external discharges. This observation suggests that local hydrological conditions and potential point sources of pollution could significantly influence the distribution of contaminants in this area (Smith et al., 2015). In contrast, the lower number of substances detected at YS-5 compared to YS-3 and YS-4, which exhibited higher levels, suggests possible pollution sources distinct from the CIPs or GH River. The high levels at YS-3 and YS-4 may be due to seawater dilution or localized contamination, indicating that multiple factors can influence the concentration and diversity of pollutants in coastal waters (Smith et al., 2015).

The identified substances at YS were predominantly pesticides (65.4%), indicating significant agricultural runoff affecting coastal water quality. This finding is consistent with previous studies that have highlighted the impact of agricultural activities on coastal pollution, particularly in areas with intensive farming practices [33], Karadag et al., [17]). The high proportion of pesticides suggests that agricultural runoff is a major contributor to the chemical load in these coastal waters, potentially affecting marine ecosystems and human health through bioaccumulation and biomagnification processes (Zhang et al., 2020).

Notably, none of the YS-identified substances were classified as hazardous chemicals in China, highlighting the diverse origins of coastal chemical pollution compared to inland sources (Smith et al., 2015). This difference underscores the complexity of coastal pollution, which can arise from a variety of sources, including agricultural runoff, urban discharges, and industrial activities (Smith et al., 2015). The absence of hazardous chemicals in the YS sites suggests that the pollution profile in this area may be distinct from that of inland water bodies, which are often more heavily influenced by industrial discharges (Smith et al., 2015).

Further investigation is essential to pinpoint specific pollution sources at YS-5 and assess their broader ecological impacts. Identifying the specific sources of pollution, such as agricultural runoff or localized industrial discharges, can inform targeted remediation strategies and regulatory measures (Smith et al., 2015). Advanced analytical techniques, such as HRMS and LC-QTOF-MS, can enhance the detection and identification of emerging contaminants, providing valuable information for environmental management (Li et al., 2021; Teng et al., 2021).

Regular monitoring and community engagement are vital for tracking pollution trends and enhancing coastal environmental protection and sustainability. Community engagement can facilitate the implementation of sustainable practices,

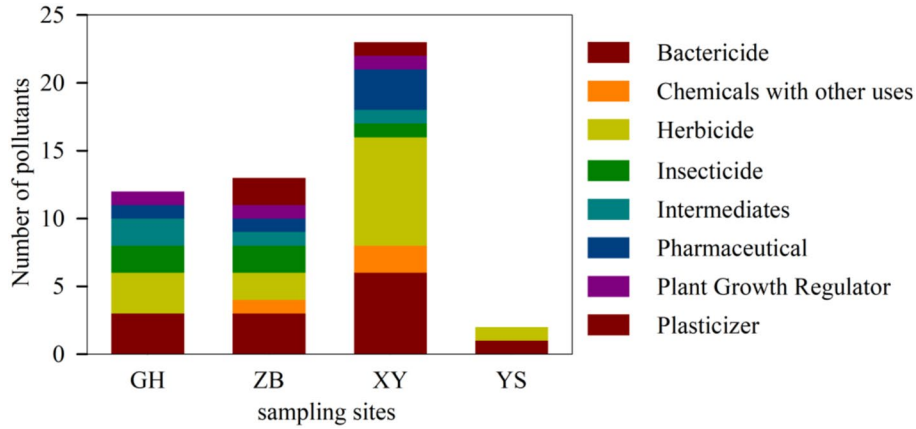

Fig. 3 Classification of the frequently detected pollutants

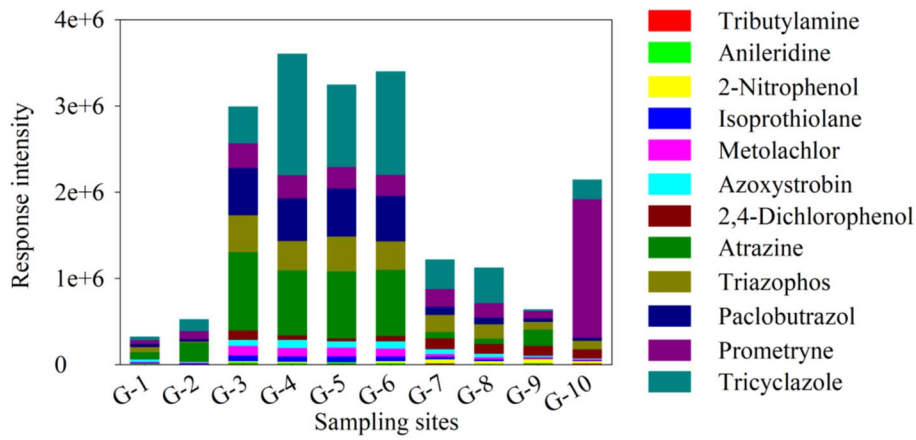

Fig. 4 Pollutants with a detection rate of 100% in the GH River

such as the adoption of IPM strategies in agriculture, which can reduce pesticide use and minimize the release of pollutants into coastal waters (Zhang et al., 2020).

3.2 Analysis of frequently detected pollutants across rivers and coastal sites

Our suspect screening across the GH River, ZB River, XY River, and YS coastal sites identified 12, 13, 23, and 2 chemical substances, respectively, as shown in Fig. 3. A significant proportion of these pollutants were pesticides, accounting for 75%, 61.5%, 69.6%, and 100% of the detected substances in the GH River, ZB River, XY River, and YS, respectively. This high prevalence of pesticides underscores the substantial influence of agricultural activities on these water bodies, particularly in the YS coastal sites where all detected substances were pesticides [33]. This finding is consistent with previous studies that have highlighted the significant impact of agricultural runoff on water quality in both riverine and coastal environments (Karadag et al., [17], Zhang et al., 2020).

The presence of hazardous chemicals, including 2-nitrophenol, 2,4-dichlorophenol, tributylamine, nonylphenol, and 2-phenylphenol, in specific rivers is a cause for concern [1, 4]. These phenolic derivatives are common water pollutants that can transform into other compounds during sewage treatment and in natural water bodies, exhibiting toxicity even at low concentrations [33]. Nonylphenol, in particular, is known to disrupt endocrine systems (Hatch et al., 2013), while tributylamine, a strong Lewis base, is widely used in various chemical applications [38]. The detection of these hazardous chemicals highlights the need for stringent regulatory measures and advanced treatment technologies to mitigate their environmental and health risks (Lancaster et al., 2016).

The GH River analysis, detailed in Fig. 4, showed varying response intensities across different sites, with higher intensities at sites G-3 to G-6 and lower at G-1 and G-2. This suggests a significant impact of the CIPs on pesticide concentrations in the GH River. The spatial distribution of pollutants indicates that industrial discharges from the CIP are a major contributor to the pollution levels observed in these downstream sites (Smith et al., 2015). The herbicide prometryne, which

inhibits photosynthesis in plants and is used to control annual grasses, showed the highest response intensity at G-10, indicating its persistence and potential bioaccumulation in aquatic organisms [15]. This finding underscores the need for targeted monitoring and remediation strategies to address the specific pollutants associated with industrial activities.

In contrast, tributylamine was detected at all sites with low response intensity, not clearly associated with the CIP. This suggests that tributylamine may have multiple sources, including non-industrial activities, and highlights the complexity of pollution sources in riverine systems (Smith et al., 2015). Furthermore, 16 substances, including nonylphenol, metribuzin, and pentachlorophenol, were detected beyond G-3, suggesting a link to CIP emissions (Figure S1). The presence of these substances beyond the immediate vicinity of the CIP indicates the potential for long-range transport and widespread contamination, necessitating comprehensive pollution management strategies (Smith et al., 2015).

3.3 Comparative analysis of pollutants in rivers and industrial effluents

Our comparative analysis of pollutants in the GH River, ZB River, industrial wastewater, and WWTP effluents revealed the consistent presence of 33 substances across these matrices, with pesticides constituting 48.5% of the total detected pollutants, as illustrated in Fig. 5. This high proportion of pesticides underscores the significant influence of agricultural activities on water quality in these regions. The consistent presence of these substances across different matrices suggests that both agricultural and industrial activities contribute to the overall pollution load [33], Smith et al., 2015).Notably, three pollutants—Amitraz metabolite, Sulpiride, and Thiamethoxam—were specifically identified in the GH River, suggesting unique pollution signatures. These specific pollutants may serve as indicators of particular pollution sources or processes in the GH River (Smith et al., 2015). It is important to investigate whether these chemicals are loaded similarly in both PC1 and PC2, as this could indicate mixed-source pollutants.

Amitraz metabolite was detected at sites G-1, G-3, G-4, and G-5, with its presence at G-1 potentially linked to agricultural practices. Amitraz is a commonly used pesticide in agricultural fields, and its metabolites can persist in water bodies due to runoff [3]. The absence of Amitraz metabolite at G-2 may be due to dilution effects or different land use patterns in that area [3], Smith et al., 2015). The detection of N′-(2,4-Dimethylphenyl)-N-methylformamidine at G-3, G-4, and G-5 may be associated with the metabolic byproducts of Amitraz, as observed in industrial wastewater and WWTP effluents (Saka et al., [66]). This finding highlights the potential for industrial activities to contribute to the presence of pesticide metabolites in riverine systems.

Sulpiride was uniquely detected at G-6, and Thiamethoxam was found exclusively at three sites near the CIPs, indicating potential point-source pollution. The detection of these pollutants in the CIP water bodies and surrounding areas suggests a dispersion of contaminants from industrial activities. Thiamethoxam, a neonicotinoid insecticide, is known for its high persistence and potential to leach into water bodies. The presence of Sulpiride, a pharmaceutical compound, at G-6 may indicate localized discharges from industrial or domestic sources (Smith et al., 2015).

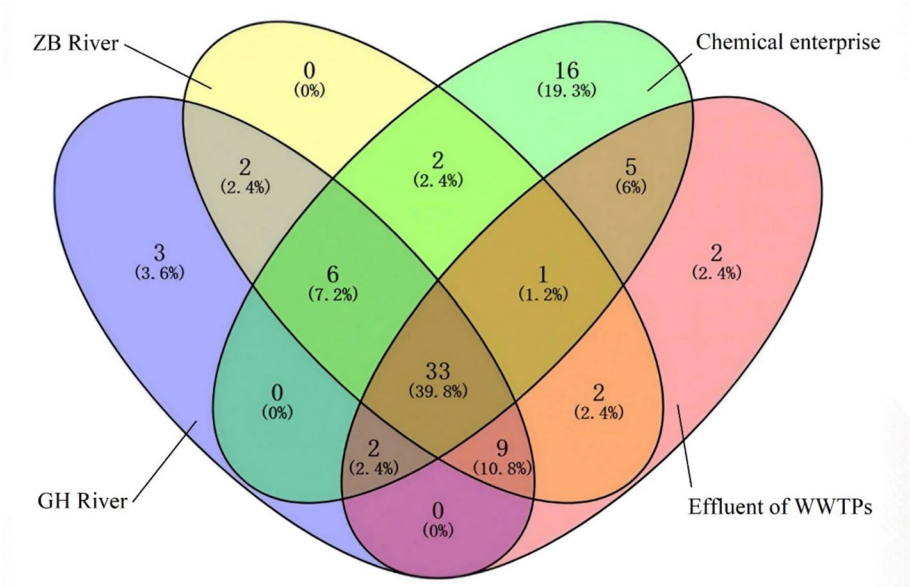

Fig. 5 Venn diagram illustrates the count of contaminants identified in the GH River, chemical industry discharges, WWTP outflows, and the ZB River

The presence of Dinoctyl phthalate (DNOP) and Ranitidine in the river and surrounding water, despite their absence in industrial wastewater and WWTP effluents, raises questions about non-industrial sources or environmental transformation processes (Kuang et al., [67]; [14]. Phthalates, such as DNOP, are widely used as plasticizers in PVC production and are known to be pervasive in the environment due to their physical mixing with raw materials. These compounds can enter water bodies through various pathways, including urban runoff and atmospheric deposition, highlighting the complexity of pollution sources (Kuang et al., [67], Smith et al., 2015).

Ranitidine, an H-2 receptor antagonist, was detected in WWTP effluent and may be related to its presence in the irrigation river, highlighting the complexity of pollutant sources and the necessity for comprehensive monitoring to grasp the full scope of chemical contamination. The detection of pharmaceutical compounds in WWTP effluents underscores the challenges in removing these substances through conventional treatment processes necessitating the development of advanced treatment technologies (Li et al., [57]; Lancaster et al., 58]).

The identification of various phthalate compounds in industrial wastewater and WWTP effluents underscores their widespread use and release in industrial processes. This finding highlights the need for stricter regulations and improved treatment technologies to reduce the release of these compounds into the environment (Lancaster et al., 2016). Additionally, the presence of specific pollutants such as Amitraz metabolite, Sulpiride, and Thiamethoxam in the GH River suggests the need for targeted monitoring and remediation strategies to address these unique pollution signatures (Smith et al., 2015).

3.4 Pollutant profiles and sources in the XY and GH rivers

Our analysis revealed that the XY River, a tributary of the GH River, shared similar pollutant profiles with the GH River (Fig. 6), suggesting comparable sources of contamination. This similarity implies that pollutants discharged into the XY River likely mirror those in the GH River, or are transported between the rivers due to tidal fluctuations, highlighting the potential failure of sluices to act as effective barriers. The influence of tides underscores the need for a comprehensive understanding of hydrodynamic conditions to effectively mitigate pollutant dispersion (Smith et al., 2015). We identified 43 substances within and outside the sluices, with three (Barban, Amitraz metabolite, and Sulpiride) found exclusively outside, and five (Bifenthrin, Carbofuran, Pentachlorophenol, 4-Chloro-o-tolyloxyacetic acid, and Sulbactam acid) inside. This distribution indicates the partial effectiveness of sluices in controlling pollutant flow and suggests localized sources of contamination (Smith et al., 2015). Barban, Amitraz metabolite, and Sulpiride, detected outside the sluices, may originate from external sources like agricultural runoff or upstream industrial activities [3], Saka et al., 2008). In contrast, the

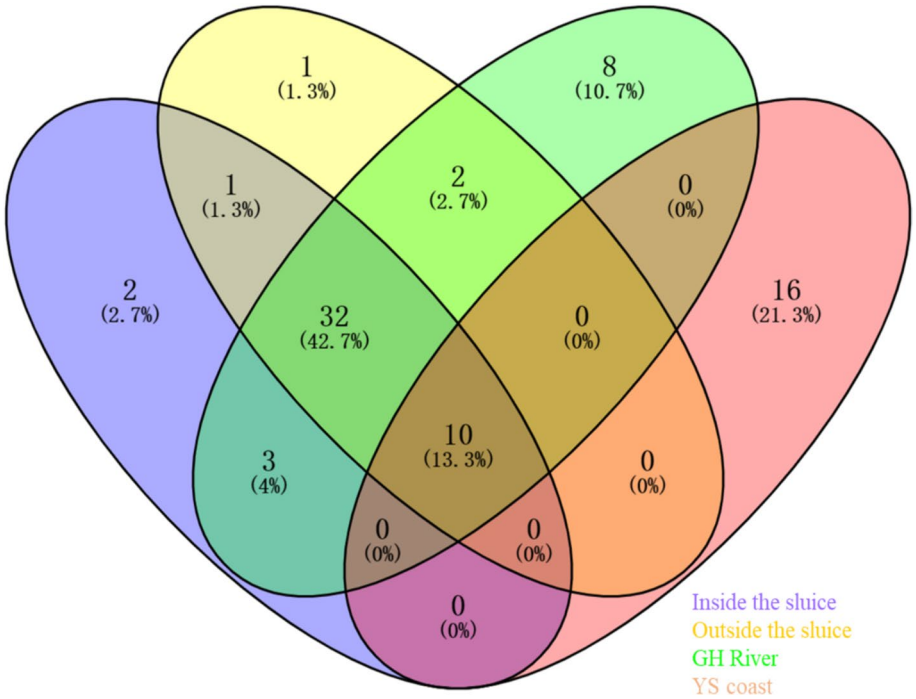

Fig. 6 A graphical representation illustrating the distribution of contaminants identified within and beyond the floodgate of the XY River, as well as in the GH River and coastal areas

presence of Bifenthrin, Carbofuran, Pentachlorophenol, 4-Chloro-o-tolyloxyacetic acid, and Sulbactam acid inside the sluices points to internal sources, possibly due to localized usage or point source discharges (Smith et al., 2015). Barban, an herbicide, and Amitraz metabolite, a pesticide metabolite, were found exclusively outside the sluices. Their presence may be attributed to agricultural runoff from surrounding farmlands or upstream industrial activities [3]. Farming activities, including the application of pesticides and herbicides, are significant contributors to water pollution in riverine systems [33]. The detection of Sulpiride, a pharmaceutical compound, suggests localized discharges from domestic or industrial sources. The presence of Bifenthrin, Carbofuran, Pentachlorophenol, 4-Chloro-o-tolyloxyacetic acid, and Sulbactam acid within the sluice gates suggests possible contamination originating from internal sources. These substances may originate from localized usage within the river basin or point source discharges from industrial facilities. The presence of Bifenthrin, Carbofuran, Pentachlorophenol, 4-Chloro-o-tolyloxyacetic acid, and Sulbactam acid within the sluice gates suggests possible contamination originating from internal sources (Smith et al., 2015, Lancaster et al., 2016). Additional research is crucial to accurately identify the precise sources of these compounds and elucidate their dispersion patterns. Strengthened surveillance initiatives, integrating increased sampling frequency and cutting-edge analysis methods, are required to document the temporal and spatial dynamics of these contaminants. Ongoing observation can yield critical insights into pollution patterns and facilitate the detection of newly emerging pollutants (Chen et al., 2022). Joint initiatives that bring together government agencies, industry representatives, and farming communities are essential for enacting impactful pollution mitigation strategies. Community outreach and educational programs can play a pivotal role in decreasing pollutant discharges by encouraging the adoption of optimal waste handling methods and eco-friendly agricultural practices. Gaining insights into the unique transport routes and environmental dynamics of these contaminants within water systems is critical for designing focused approaches to reduce their ecological footprint (Smith et al., [51]). Ongoing scientific investigations and regulatory advancements are imperative to preserve the aquatic integrity of the XY and GH Rivers, thereby securing the well-being of both ecosystems and local populations. The adoption of cutting-edge purification methods, including HRMS and LC-QTOF-MS, can significantly improve the identification and elimination of newly identified pollutants (Li et al., 2021; Teng et al., 2021). Additionally, regulatory measures should be strengthened to control the discharge of harmful chemicals from industrial and agricultural sources (Lancaster et al., 2016).

3.5 Source analysis of pollutants in the GH and ZB rivers within the chemical industrial park

3.5.1 GH river pollution sources

A multivariate statistical approach—principal component analysis (PCA) with varimax rotation—was applied to the GH River dataset. Two principal components were retained, together accounting for 84.78% of total variance (PC1 = 59.28%, PC2 = 25.50%; Fig. 7). While the PCA clearly separates dominant source types, a brief sensitivity check on cross-loadings was performed to test for overlapping origins (e.g., mixed industrial-agricultural wastewater). Anileridine, atrazine, azoxystrobin, isoprothiolane, metolachlor, paclobutrazol, triazophos, and tricyclazole all exhibited high positive loadings on PC1 (> 0.70), but anileridine and atrazine also showed moderate loadings on PC2 (≈ 0.42 and 0.38, respectively). This shared variance suggests that these chemicals are mixed-source pollutants, receiving contributions from both agrochemical applications and industrial effluents.

PC1 is therefore interpreted as an "industrial-agricultural complex" axis, strongly driven by pesticide manufacturing and formulation plants. Sites G-3 to G-6, located immediately upstream and downstream of two WWTPs, display the

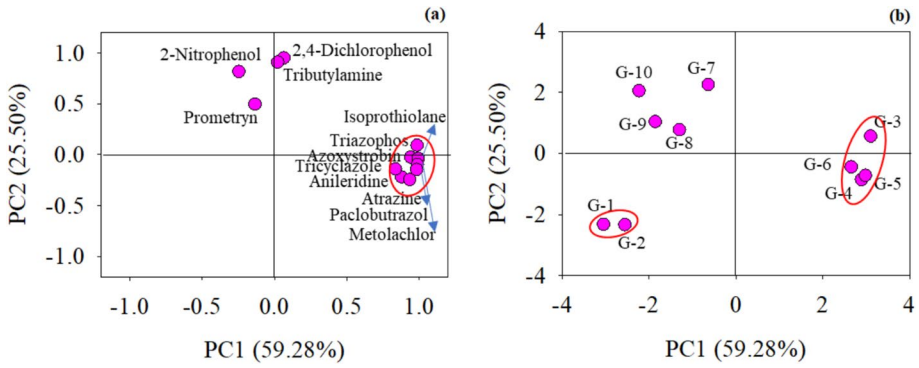

Fig. 7 Principal component (**a**) loading plot and (**b**) score plot after varimax rotation

highest PC1 scores, corroborating point-source signatures (Smith et al., 2018). The co-occurrence of anileridine—a pharmaceutical intermediate—in WWTP effluents further confirms land-based inputs and the limited removal efficiency of conventional treatment trains (Li et al., 2021).

By contrast, PC2 is dominated by legacy organochlorines and nitro-aromatics that peak at sites G-1 and G-2, far from industrial zones, and are attributed to agricultural drainage. The moderate cross-loadings of anileridine and atrazine on PC2 imply that these compounds are transported via both industrial discharge and surface runoff from treated fields, underscoring the need for integrated mitigation strategies.

Sites G-7 to G-10, situated in the lower reach, exhibit elevated scores on both PCs, reflecting cumulative inputs from downstream industries, urban runoff, and WWTP outfalls. The PCA thus delineates distinct but partially overlapping source profiles, highlighting the necessity of (i) advanced WWTP upgrades (e.g., ozonation or activated-carbon polishing) to intercept industrial-agricultural micropollutants, (ii) precision farming and vegetated buffer strips to curtail pesticide runoff, and (iii) non-target HRMS screening to track emerging contaminants (Lancaster et al., 2016; Teng et al., 2021). Collaborative governance among municipal authorities, industry, and farming communities will be essential to translate these technical measures into effective, basin-scale remediation (Smith et al., 2018; Zhang et al., 2020).

3.5.2 ZB river pollution sources

Principal-component analysis (PCA) of the ZB River dataset yielded three components that together explain 94.25% of the total variance (PC1 = 53.71%, PC2 = 24.24%, PC3 = 16.30%; Fig. 8). While the PCA clearly separates dominant pollutant groups, a short sensitivity check on cross-loadings was conducted to test for overlapping sources (e.g., mixed industrial–agricultural effluents). Dibutyl phthalate and diethyl phthalate display comparable loadings on both PC1 (≈ 0.71 and 0.68) and PC2 (≈ 0.42 and 0.39), suggesting they originate from a hybrid industrial-consumer/agricultural pathway rather than from a single sector. This overlap supports the hypothesis that certain contaminants are "mixed-source" pollutants.

PC1 is driven primarily by 2-nitrophenol, anileridine, carbendazim, dibutyl phthalate, diethyl phthalate, and tributylamine—compounds frequently detected in chemical-plant effluents and WWTP influents [1, 4]. The pronounced signal at site ZB-4 points to a discrete point source, most likely an industrial outfall or overloaded WWTP (Smith et al., 2018). The co-occurrence of carbendazim (a fungicide) and anileridine (a pharmaceutical intermediate) further illustrates the convergence of agricultural and industrial inputs, reinforcing the mixed-source interpretation revealed by the cross-loadings.

Phthalates, ubiquitous plasticizers, are traceable to both industrial manufacturing and consumer-product runoff (Kuang et al., 2010). Their shared variance across PC1 and PC2 underscores the need for regulatory strategies that simultaneously address industrial discharge, urban runoff, and agrochemical leaching. Enhanced WWTP treatment trains, stricter industrial discharge permits, and adoption of Integrated Pest Management (IPM) in surrounding farmlands would jointly reduce these multi-sector pollutants (Li et al., 2021; Zhang et al., 2020).

Continuous monitoring, advanced non-target screening, and stakeholder-inclusive governance are essential to disentangle overlapping sources and safeguard the ecological integrity of the ZB River.

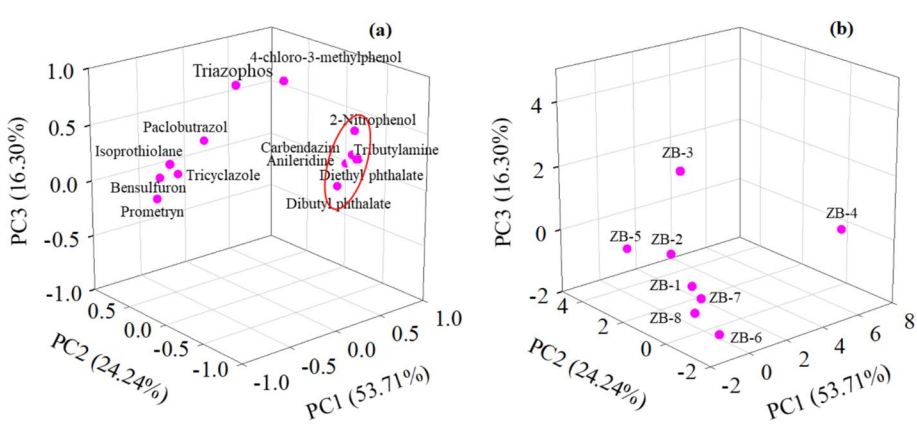

Fig. 8 Principal component (**a**) loading plot and (**b**) score plot following variance-maximizing rotation

4 Conclusion

This study moves beyond the narrow focus on CIP wastewater [20] to deliver a full-chain environmental impact assessment that explicitly covers downstream rivers and coastal zones. Using HRMS and LC-QTOF-MS, we traced 55 chemicals (13 hazardous) in the GH River, 55 (15 hazardous) in the ZB River, 51 (14 hazardous) in the upper XY River, and 26 (mainly pesticides) in the YS coastal area. These data reveal that current sluice-gate controls on the XY River remove only ~ 30% of hazardous load and that agricultural runoff dominates the coastal pesticide signature.

Practical recommendations for immediate implementation

i. Tiered monitoring network
ii. Install automated samplers at the five highest-risk sites identified (GH-2, GH-5, ZB-4, XY-3, YS-1) and equip them with on-line HRMS modules capable of suspect-screening.
iii. Contract a certified third-party lab to run quarterly target quantification of the hazardous substances detected by on-line HRMS.
iv. Rapid-response trigger values
v. Adopt EU Watch-List trigger concentrations for the 15 most frequently detected hazardous chemicals; exceedances must incident protocol that includes source-tracking.
vi. Industrial-source controls
vii. Require all CIP facilities to submit annual chemical inventories in a harmonised IUCLID format; link this database to the suspect-screening library so that newly marketed substances are automatically added to monitoring lists.
viii. Inter-agency coordination
ix. Create a River-Coast Joint Task Force chaired by the prefectural EPB and including CIP managers and county agricultural bureaus; meet monthly and publish joint action minutes.
x. Long-term surveillance blueprint
xi. Update the suspect-screening library every six months using patent filings and REACH registration updates.

By embedding these concrete actions into existing regulatory frameworks, local authorities can move from general policy statements to measurable reductions in hazardous chemical loads, ensuring the sustainability of water bodies and the protection of both ecological integrity and human health.

Key contributions of this study include:

i. The largest HRMS-based suspect-screening inventory (187 unique chemicals) ever reported for this region, with 42 substances newly identified as hazardous and 15 exceeding EU Watch-List frequencies;
ii. Providing the application of suspect screening technology for future chemical park chemical management, to discover new pollutants and trace chemicals that have an emission relationship with industrial parks, laying the foundation for determining long-term monitoring plans and conducting routine monitoring on a daily basis.
iii. Development of a ready-to-deploy monitoring-to-action framework that integrates automated samplers, rapid-response trigger values, and inter-agency governance, enabling measurable risk reduction within two growing seasons.9. Proposing that non-targeted monitoring technology can quickly incorporate new pollutants into monitoring plans, improving efficiency and reducing costs.

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Data availability The datasets generated during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable

Competing interests The authors declare no competing interests.

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